Downstream Utility of Synthetic Data for Clustering

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Abstract

Synthetic data are released as privacy-preserving substitutes for confidential records, but their value depends on whether analyses of the release reproduce the original ones. We evaluate this for cluster analysis, whose output depends on point-cloud geometry. In a factorial simulation of 144 scenarios crossing separation, number of groups, dimensionality, correlation and distribution family (720,000 evaluations), we compare k-means and Ward's linkage on data synthesised by sequential CART. Agreement between cluster counts estimated from synthetic and original data rises from 0.75 to 0.98 with separation and depends little on the number of groups. Against the planted number of groups the two algorithms perform alike, but against the original data k-means is more stable (0.864 versus 0.823) and shows less geometric distortion. CART is close to neutral for Normal components but lowers

agreement for skewed Gamma components, especially for Ward's linkage. Utility must be validated against the intended analysis.

Keywords: Synthetic data, Cluster analysis, Statistical disclosure control, Analytical utility, Simulation study

1 Introduction

Synthetic data are artificial records produced by a generative model fitted to a confidential dataset, released in place of the original records so that the statistical structure of the data is retained while no real individual is exposed. The idea originates in statistical disclosure control, where Rubin (1993) proposed treating the confidential values as missing and releasing multiply-imputed replacements, and Little (1993) developed the companion analysis of masked data; Drechsler (2011) gives a book-length treatment of the resulting framework, and Matthews and Harel (2011) survey the wider family of disclosure-limitation methods within which synthesis sits. Synthetic data were initially developed mainly for statistical agencies, but their application has since expanded to many other fields. General-purpose synthesisers are now distributed as statistical software (Nowok, Raab, and Dibben, 2016; Templ, Meindl, Kowarik, and Dupriez, 2017), and administrative-data research is an active area of application (Kokosi, De Stavola, Mitra, Frayling, Doherty, Dove, Sonnenberg, and Harron, 2022). Synthetic records are also generated for health services research (Gonzales, Guruswamy, and Smith, 2023; Giuffrè and Shung, 2023), for biobehavioural science, where they additionally serve reproducibility and hypothesis generation (Quintana, 2020), and, increasingly, through deep generative architectures applied to tabular data (Goyal and Mahmoud, 2024; Zhao, Kunar, Birke, Van der Scheer, and Chen, 2024).

Synthetic data are mainly generated to support subsequent analyses, and cluster analysis is one of the most widely used unsupervised approaches: given unlabelled observations, it partitions them into groups that are internally homogeneous and mutually distinct. Two main approaches have dominated applied practice for decades: centroid-based methods, such as k-means (MacQueen, 1967; Steinley, 2006), which minimise the within-cluster sum of squares and remain the most widely deployed clustering algorithms (Ahmed, Seraj, and Islam, 2020); and agglomerative hierarchical methods, such as Ward's linkage (Ward, 1963), which build a dendrogram by successive merging and do not require the number of groups to be fixed in advance. Both remain central to contemporary applications: in single-cell transcriptomics, where partitions define cell types (Zappia and Oshlack, 2018), and in mass cytometry, where the choice among clustering methods materially changes the biological conclusion (Liu, Song, Wong, Zhang, Yu, Lin, and Ding, 2019).

These two areas intersect whenever a clustering analysis must be performed on data that cannot be released in raw form. In clinical and administrative settings, for example, subgroup discovery can be highly valuable, but access to the original data is often restricted for privacy reasons (Gonçalves, Ray, Soper, Stevens, Coyle, and Sales, 2020; El Emam, Mosquera, and Bass, 2020; Bhanot, Qi, Erickson, Guyon, and Bennett, 2021). In model development, access to the original data can also be slow or limited, and synthetic records then serve as an augmentation and development resource (Rankin, Black, Bond, Wallace, Mulvenna, and Epelde, 2020). Synthetic data can therefore provide a practical solution in both situations. However, when an analysis is performed on synthetic rather than original data, it is important to assess whether the resulting partition is similar to the one that would have been obtained from the original records. This is what the *analytical utility* of synthetic data refers to.

Analytical utility can be assessed from a general and from a specific perspective. *General* utility asks whether the synthetic data resemble the original in a global

sense, and *specific* utility asks whether a particular analysis yields matching conclusions (Snoke, Raab, Nowok, Dibben, and Slavkovic, 2018). The distinction matters because the two can diverge: a generator that reproduces marginal distributions and low-order correlations faithfully may still distort the higher-order geometry on which a specific procedure depends. Empirical work bears this out. Benchmarking studies of synthetic health records find that utility rankings depend heavily on which downstream task is used to measure them (Yan, Yan, Wan, Zhang, Omberg, Guinney, Mooney, and Malin, 2022; El Emam, Mosquera, Fang, and El-Hussuna, 2022), and direct replications of published analyses on synthetic counterparts show agreement that varies by study and by estimand (Reiner Benaim, Almog, Gorelik, Hochberg et al., 2020). A recent systematic review of tabular synthesis tools makes the same point from the measurement side, finding no consensus on which utility metrics should be reported and little agreement between them (Lautrup, Hyrup, Zimek, and Schneider-Kamp, 2024). Utility is therefore not a scalar property of a dataset; it is a property of the pairing between a dataset and the analysis applied to it.

Clustering provides a useful approach for assessing this relationship. Unlike a regression coefficient, which summarises a conditional mean and is comparatively robust to local perturbation, a partition is sensitive to the fine structure of the point cloud: relative positions, local density, separability and within-group dispersion. Two datasets can share identical means, variances and pairwise correlations and still yield entirely different partitions. Distortion that a generator introduces into local geometry may therefore appear in a clustering analysis before it appears in a resemblance check.

This article asks whether CART-based sequential synthesis, the default nonparametric engine of `synthpop` (Nowok et al., 2016) and a widely used method for tabular data, preserves the structure that clustering relies on. This study extends the line of inquiry opened by Chen (2025), which established a simulation harness for this question but restricted attention to k-means on Normal data and relied on

a resemblance statistic that proved insensitive to structural quality. We broaden the scope in three directions: a second, structurally different clustering method; a skewed distribution family that exposes the axis-aligned behaviour of tree-based synthesis; and a set of structural metrics that compare the partition recovered from the synthetic data with the partition recovered from the original data.

The utility of synthetic data for clustering analysis is assessed through three main objectives. First, to determine whether CART-synthesised data preserve the structure needed to recover the number of clusters, and how this depends on separation, number of groups, dimensionality, feature correlation and distribution family. Second, to determine whether the cluster structure is preserved beyond the number of clusters: how much fidelity of centroid location, within-group dispersion and group sizes is lost under synthesis, and which design factors drive that loss. Third, to determine whether partitioning and hierarchical methods respond similarly to synthetic distortion, or whether one is systematically more robust than the other. We address these questions with a factorial simulation. For each scenario, several original datasets are simulated from a mixture with known group structure, and each original dataset is then synthesised many times with CART. We compare the number of clusters recovered from each synthetic dataset with two references: the true number of groups used to simulate the original dataset, and the number of clusters recovered from that original dataset by the same algorithm.

2 Methods

The study is a controlled factorial simulation. We first state the discrepancy framework and the two clustering algorithms, then the generation of original and synthetic data, then the metrics used to quantify structural degradation, and finally the way those metrics are summarised across replications.

2.1 A discrepancy framework for structural utility

Let D_R denote the original (reference) data and G a generative mechanism, so that $D_S = G(D_R)$ is the synthetic dataset. Let A be the downstream algorithm, producing summaries $\theta_R = A(D_R)$ and $\theta_S = A(D_S)$. Utility is quantified by a discrepancy

$$\Delta(\theta_R, \theta_S) = f(N, p, \rho, \sigma, k, F), \tag{1}$$

studied as a function of the sample size N, dimensionality p, feature correlation ρ, separation σ, number of latent groups k and distribution family F. A generator is useful for a task over a region of the design space when Δ remains small throughout it; the scientific content of an evaluation is the shape of Δ: which factors inflate it, which leave it unchanged, and whether different algorithms respond similarly.

In the clustering setting, θ comprises the estimated number of groups $\hat{k}$ together with the geometry of the recovered partition: its centroids, its group sizes and its within-group dispersion. Accordingly Δ is instantiated as a family of structural-fidelity measures defined in Section 2.4.

To isolate how synthesis affects cluster structure, we use simulations with known ground truth rather than real administrative data. Each scenario fixes a configuration $(N, p, k, \rho, \sigma, F)$ for which the true group membership of every point is known by construction. This makes it possible to separate the difficulty of the clustering problem itself from the distortion introduced by synthesis, avoids relying solely on resemblance-based evaluations, and allows Δ in Eq. (1) to be traced as an explicit function of each design factor.

2.2 Clustering algorithms

We compare two standard approaches: k-means (MacQueen, 1967) as a partitioning method, and agglomerative clustering with Ward's linkage (Ward, 1963) as a hierarchical method. They are chosen because they rely on different optimisation strategies and may therefore respond differently to the distortions introduced by tree-based

synthesis (described in Section 2.3.2). k-means minimises the within-cluster sum of squares and so tends to favour convex, isotropic groups of comparable size (Steinley, 2006; Xiong, Wu, and Chen, 2009), with decision boundaries given by the bisectors of the centroid Voronoi tessellation; we retain the lowest-inertia solution over ten random initialisations. Ward's linkage merges, at each step, the pair of clusters whose union least increases the total within-cluster error sum of squares. The two methods therefore share a variance-based criterion, but Ward's merges are greedy and never revisited, so a merge driven by a local density irregularity early in the hierarchy propagates to the final partition, whereas k-means reassigns points iteratively; the two families have long been known to differ in their sensitivity to perturbations of the data (Milligan, 1980). Appendix A states both objectives formally.

2.3 Data generation

2.3.1 Original data

Each original dataset D_R contains $N = 1000$ observations drawn from a k-component mixture in $\mathbb{R}^p$ with known group labels and equal component sizes (N/k observations each, up to rounding). The minimum pairwise centroid distance is controlled by the separation σ: the first centroid sits at the origin and each subsequent centroid is drawn uniformly on the sphere of radius σ about the origin and accepted only if it lies at distance at least σ from all centroids already placed. Within-component covariance equals a correlation matrix with unit diagonal and off-diagonal entries ρ, giving uncorrelated (spherical) components at $\rho = 0$ and correlated (elliptical) components at $\rho = 0.4$; separation is therefore expressed in units of the within-component standard deviation. In the *Normal* family, components are multivariate Gaussian with the specified centroid and covariance. In the *Gamma* family, independent Gamma variates (shape $= 1.5$, rate $= 5$) are standardised to zero mean and unit variance within each component, correlated using the Cholesky factor of the target correlation matrix (Johnson, 1987), and shifted to the component

centroid. The two families thus share their first two moments within components and differ only in shape: before correlation is induced, each Gamma margin has skewness $2/\sqrt{1.5} \approx 1.63$, producing right-skewed groups that stress the axis-aligned nature of CART.

The factors we vary are the same high-level geometric properties used to parameterise cluster-analysis benchmarks generally, namely overlap between groups, their number, the ambient dimension and the shape of the components (Qiu and Joe, 2006; Zellinger and Bühlmann, 2025); here they serve not to benchmark clustering algorithms against each other but to locate the regions in which synthesis preserves the structure those algorithms depend on.

Table 1 lists the design factors. Each factor is varied to examine how it affects the conditions under which synthesis succeeds or fails: crossing all five gives $3 \times 4 \times 3 \times 2 \times 2 = 144$ scenarios. For each scenario, $n = 5$ independent original datasets are simulated, and each original dataset is synthesised $m = 1000$ times, resulting in $144 \times 5 \times 1000 = 720{,}000$ synthetic-versus-original evaluations.

2.3.2 Synthetic data (CART)

For every original dataset, synthetic data $D_S = G(D_R)$ are generated with the CART engine of `synthpop` (Nowok et al., 2016), using proper synthesis (the trees are fitted to a bootstrap sample of the original data) and a minimum leaf size of ten. Variables are synthesised sequentially in the order $X_1, \ldots, X_p$. The first variable is sampled from its observed marginal distribution. Each subsequent variable X_j is then modelled by a regression tree (Breiman, Friedman, Olshen, and Stone, 1984) on the already-synthesised variables, and synthetic values are drawn from the empirical distribution of the observed values in the corresponding leaf,

$$X_j^{\text{syn}} \sim \widehat{P}\big(X_j \mid \text{leaf}(X_1^{\text{syn}}, \ldots, X_{j-1}^{\text{syn}})\big). \tag{2}$$

Table 1 Factors of the simulation design and the question each addresses. Crossing all five gives 144 scenarios; with $n = 5$ independent original datasets per scenario and $m = 1000$ synthetic datasets generated from each, the design yields 720,000 evaluations.

Factor	Levels	Rationale
Separation σ	0.1, 2, 6, 10	Spans overlapping to clearly distinct groups; tests how much separation between groups is needed before synthesis can preserve the structure.
Number of groups k	2, 3, 4	More groups increase the complexity of the cluster structure that the generator needs to preserve.
Dimensionality p	2, 5, 10	Each synthesised dimension compounds the axis-aligned leaf structure of the generator.
Correlation ρ	0, 0.4	Tests whether dependence between features degrades a variable-by-variable synthesis.
Distribution family F	Normal, Gamma	Contrasts symmetric Gaussian density with skewed Gamma density.
Sample size N	1000 (fixed)	Held constant so that variation is attributable to the factors above.

This makes the method flexible and distribution-free, but it imposes an axis-aligned, piecewise-constant structure on the joint distribution: the synthetic point cloud is assembled from rectangular blocks whose boundaries are the tree splits, and synthetic values are resampled observed values. When the original data are smooth and symmetric, these blocks tile the space finely enough to be barely visible; when the data are skewed or heavy-tailed, they can become visible as a hardening of smooth density gradients into discrete steps. Detecting this failure mode, and measuring how strongly it propagates into the recovered partition, is the purpose of the evaluation. The group label is appended as the last variable of the synthesis sequence, so it does not influence the synthesis of the features; it is never used by the clustering algorithms.

Synthesis was carried out with `synthpop` 1.9.2 (Nowok et al., 2016) under R 4.3 (R Core Team, 2024), and clustering and evaluation with `scikit-learn` (Pedregosa et al., 2011) in Python. Both stages are embarrassingly parallel over scenarios and were distributed across 18 worker processes on a single multi-core workstation (Appendix B).

2.4 Evaluation metrics

Structural utility is quantified by two count-agreement measures and three geometric-fidelity measures. All variables are standardised before clustering. The scaler is fit on the original data and applied unchanged to the corresponding synthetic dataset, so that all geometric quantities are expressed in a common coordinate system; standardising each dataset separately would rescale away part of the very distortion the metrics are meant to detect (see Section 4.2).

Number of groups.

For a dataset and an algorithm, the number of groups is estimated by maximising the average silhouette width (Rousseeuw, 1987) over candidates $\{2, \ldots, k_{\max}\}$, with $k_{\max} = \min(k + 3, 8)$, the rule inherited from Chen (2025). Because this rule ties the search range to the planted k, every count analysis is repeated with the fixed range $\{2, \ldots, 8\}$ (Supplementary Section S3). We write $\hat{k}_R$ for the estimate obtained from an original dataset and $\hat{k}_{S,m}$ for the estimate obtained from the m-th synthetic dataset generated from it, and $\hat{k}_{R,m}$ for the estimate from the original dataset that generated synthetic dataset m. For a configuration with M evaluations we report two agreement proportions,

$$\mathrm{RR} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\big[\hat{k}_{S,m} = k\big], \qquad \mathrm{AG} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\big[\hat{k}_{S,m} = \hat{k}_{R,m}\big], \tag{3}$$

where $\mathbf{1}[\cdot]$ is the indicator function. The recovery rate RR compares the synthetic estimate with the planted number of groups. It therefore confounds two sources of

failure: distortion introduced by synthesis, and the inability of the clustering procedure to recover the planted structure even from the original data (for instance, when the groups overlap so strongly that a single diffuse cloud is a reasonable description). The agreement AG compares the synthetic estimate with the estimate from the original data, obtained with the same algorithm and the same selection rule, and thus isolates the effect of synthesis. AG is the primary measure of this study; RR, together with the recovery rate of the original data itself, $\mathbf{1}[\hat{k}_R = k]$, is reported to separate the two sources of failure.

Geometric fidelity.

The geometric metrics are computed with the number of clusters fixed at the planted k on both sides, so that they describe the geometry of the partition conditional on the correct count. They require the groups recovered in D_S to be matched to those in D_R, since cluster labels are arbitrary. We build the matrix of pairwise centroid distances between the k original and k synthetic clusters and solve the resulting linear assignment problem (Kuhn, 1955), obtaining the one-to-one matching that minimises total centroid distance. All fidelity metrics are computed on the aligned clusters.

Group-balance error (ΔG).

This metric evaluates whether the distribution of observations across clusters is preserved under synthesis. It is defined as

$$\Delta G = \left| G_S - G_R \right|, \tag{4}$$

where G_R and G_S are the Gini coefficients of the cluster-size vectors of the partitions recovered from the original and synthetic data. For sizes $\{n_1, \ldots, n_k\}$,

$$G = \frac{\sum_{a=1}^{k} \sum_{b=1}^{k} |n_a - n_b|}{2k \sum_{a=1}^{k} n_a}, \tag{5}$$

with $G = 0$ for perfectly equal groups. A value of ΔG close to 0 indicates that synthesis has preserved the distribution of cluster sizes. Because the planted groups have equal sizes, G_R reflects only the imbalance that the algorithm itself produces on the original data.

Mean centroid distance (MCD).

This metric measures the average distance between the centroids of the original clusters and the centroids of the corresponding synthetic clusters. With aligned original and synthetic centroids μ_j^R and μ_j^S,

$$\text{MCD} = \frac{1}{k}\sum_{j=1}^{k}\left\|\mu_j^R - \mu_j^S\right\|, \tag{6}$$

so small values indicate that synthesis preserves the location of the groups.

Mean variance difference (MVD).

This metric evaluates whether the synthetic clusters retain a dispersion similar to that of the original clusters. Writing v_j^R and v_j^S for the mean squared distance of points to their centroid in aligned original and synthetic clusters,

$$\text{MVD} = \frac{1}{k}\sum_{j=1}^{k}\left|v_j^R - v_j^S\right|. \tag{7}$$

2.5 Statistical analysis

The design is hierarchical: $m = 1000$ synthetic datasets are nested within each of $n = 5$ original datasets, which are nested within each of the 144 scenarios. Agreement proportions are reported directly over evaluations. Fidelity metrics are aggregated first to a scenario mean and then averaged across scenarios, with dispersion reported as a 95% interval (mean ± 1.96 standard errors) computed from the scenario-level means; using the scenario as the unit of replication avoids treating the m synthetic draws from one original dataset as independent. Algorithm comparisons use paired

Wilcoxon signed-rank tests across the 144 scenario-level means. To rank the design factors, we fit a main-effects linear model to the 720 replicate-level agreement proportions (one per original dataset) and report partial η^2 for each factor. As a reference for how much disagreement ordinary sampling variability alone would produce, we also compute, within each scenario, the proportion of pairs of independent original datasets that yield the same $\hat{k}_R$ (Supplementary Section S4).

For visualisation we use the two-dimensional scenarios, plotting the standardised features against each other; no dimensionality reduction is applied, so the panels show the geometry the algorithms actually operate on. In the paired original/synthetic displays, cluster labels are matched by the same Hungarian alignment used for the fidelity metrics and then oriented so that colour tracks vertical position consistently across panels.

3 Results

3.1 The block-like artefact

Figures 1 and 2 show a two-dimensional, two-group scenario at moderate separation ($\sigma = 2$): the original data in the top row and one CART synthetic replicate below, each partitioned by both algorithms. Reading across a row shows how the two algorithms divide the same cloud; reading down a column shows what synthesis did to that division.

On Normal data (Fig. 1) the overall cluster structure is largely preserved after synthesis. Because CART resamples observed values within leaves, the synthetic cloud contains repeated coordinates, visible as short vertical and horizontal runs of points, most clearly near the edges of the range. Both algorithms divide the synthetic cloud along a boundary of similar orientation to the one they find in the original, although the Ward partition of the synthetic data assigns a visibly smaller share of points to the upper group. At $\sigma = 2$ the true groups overlap substantially. Neither algorithm

reproduces the true labelling exactly: where the true groups interpenetrate, both algorithms replace them with a sharp linear boundary through the overlap region.

On Gamma data (Fig. 2) the block-like artefact introduced by CART becomes more visible. The original groups are right-skewed, with a dense corner and long tails running out along both axes. In the synthetic cloud those tails break into discrete, axis-aligned patches, and the dense corner is reproduced as a set of rectangular blocks rather than a continuous gradient. This is the characteristic artefact of sequential tree synthesis: the piecewise-constant leaf structure of Eq. (2) can become more apparent for skewed, non-spherical densities. The partitions still divide the cloud in roughly the same orientation, but the boundary is displaced and the resulting groups have different internal spread, the distortion that the fidelity metrics quantify below.

3.2 Number of groups

The number of clusters estimated from synthetic data can be evaluated against two references, each addressing a different question (Section 2.4). Comparison with the planted number of groups (RR) assesses whether the synthetic data support recovery of the underlying truth, and so combines synthesis distortion with the inability of the clustering procedure to find the planted structure at all. Comparison with the number estimated from the original data (AG) isolates the effect of synthesis. Both are reported throughout, and Figure 3 shows each against separation, stratified by the true number of groups.

Separation is the decisive factor under both references. Against the planted number of groups, recovery rises from 0.272 (k-means) and 0.296 (Ward) at $\sigma = 0.1$, through 0.304 and 0.307 at $\sigma = 2$, to 0.810 and 0.789 at $\sigma = 6$ and 0.954 and 0.950 at $\sigma = 10$. The transition is sharp: up to $\sigma = 2$ recovery is low and dominated by the overlap of the groups themselves, while at $\sigma = 10$ it is near-perfect for every k.

Against the number estimated from the original data the picture is markedly more favourable, and the gap is largest exactly where the groups overlap: agreement

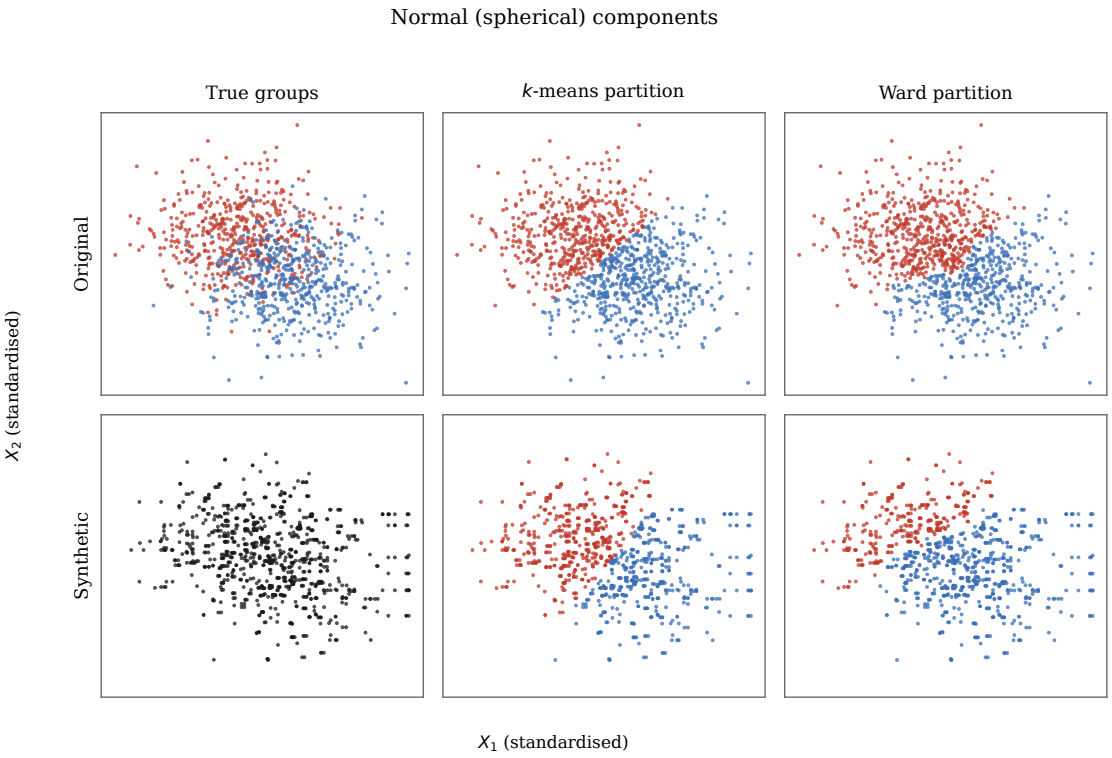


Fig. 1 Normal (spherical) components: geometry largely preserved under synthesis. Rows are the original data (top) and one CART synthetic replicate (bottom); columns are the true group labels, the k-means partition and the Ward partition. In the two partition columns, colours are matched across rows by Hungarian alignment, so the same colour denotes corresponding groups in the original and synthetic panels. The lower-left panel is drawn in a single colour: the group label in the synthetic data is itself synthesised, so it is a prediction from the synthetic features rather than a planted truth, and it has no bijective correspondence with the original groups. Both rows contain the same number of observations ($N = 1000$); the synthetic panels can look sparser only because CART resamples observed values, so that points coincide more often. Scenario: $k = 2$, $p = 2$, $\rho = 0$, $\sigma = 2$.

is 0.751 (k-means) and 0.676 (Ward) at $\sigma = 0.1$, rising to 0.983 and 0.976 at $\sigma = 10$, for overall agreement of 0.864 and 0.823. The explanation is that at low separation the algorithms frequently fail to recover the planted count from the original data as well: at $\sigma = 0.1$, k-means finds the true k in only 0.289 of original datasets and

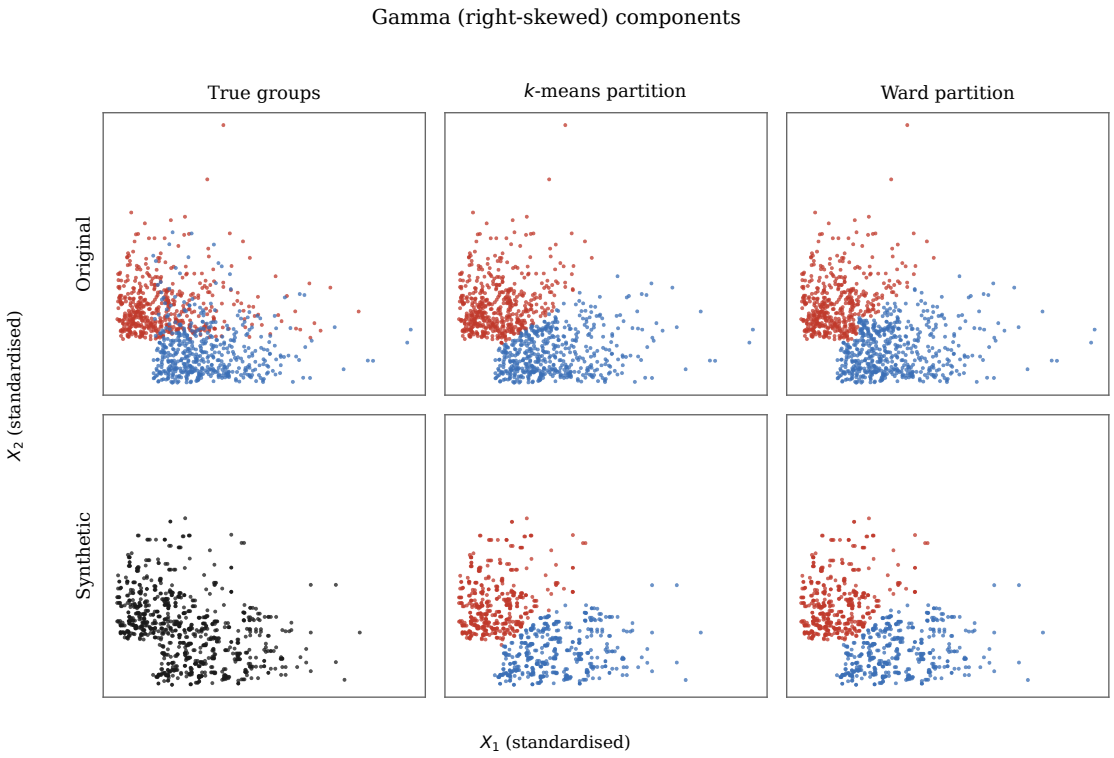


Fig. 2 Gamma (right-skewed) components: the same synthesis mechanism breaks smooth tails into axis-aligned blocks. Panel layout, colour matching, the single-colour lower-left panel and the scenario are as in Fig. 1, and both rows again contain $N = 1000$ observations. Comparing the lower-left panel with the upper-left one shows the artefact directly: the continuous density gradient of the original becomes a set of discrete rectangular patches.

Ward in 0.306. Much of what the first reference records as a failure of synthesis is therefore a failure of clustering under overlap, which synthesis reproduces faithfully. As a reference point for the size of this agreement, two independent original datasets from the same scenario yield the same $\hat{k}_R$ in 0.816 (k-means) and 0.772 (Ward) of pairs; a synthetic dataset thus agrees with its parent more often than a fresh sample from the same population would, except in the most overlapping regime ($\sigma = 0.1$), where the two are comparable for Ward and synthetic agreement is lower for k-means (Supplementary Section S4).

The number of groups behaves very differently under the two references. Against the planted count, recovery is much higher for fewer groups: 0.843 for k-means at $k = 2$, against 0.473 at $k = 3$ and 0.439 at $k = 4$; in the overlapping regime the three- and four-group cases are almost never recovered, yet they exceed 0.90 once $\sigma = 10$. Against the original data, this pattern largely disappears (0.885, 0.863 and 0.844). The difficulty of larger k thus lies mainly in the clustering problem, which synthesis reproduces, rather than in synthesis itself. The partial η^2 values from the main-effects model make the contrast explicit: for the planted reference, separation (0.55 for k-means, 0.57 for Ward) and k (0.31 and 0.37) dominate, whereas for agreement with the original data separation (0.20 and 0.21) is followed by correlation (0.10 and 0.07) and distribution family (0.03 and 0.10), with k below 0.01. Table 2 reports the marginal means for every design factor under both references.

Two further patterns in Table 2 deserve comment. Agreement with the original data is higher for correlated components (0.929 versus 0.800 for k-means) and does not decline with dimensionality. The second of these should not be read as evidence that synthesis is easier in higher dimensions: at $p = 10$ and $\sigma \leq 2$ the silhouette criterion selects $\hat{k}_R = 2$ for 87–90% of original datasets (k-means), so original and synthetic estimates coincide largely because both collapse onto the smallest candidate. The distribution family produces a consistent gap under both references, which Section 3.4 examines.

3.3 Geometric fidelity

Figure 4 traces two of the geometric-fidelity metrics against separation. Mean centroid distance falls steeply as groups separate, from 0.60 (k-means) and 0.86 (Ward) at $\sigma = 0.1$ to 0.09 and 0.10 at $\sigma = 10$, because well-separated groups pin their centroids robustly regardless of the local block artefact. Mean variance difference also declines, more gradually, from 0.35 and 0.61 to 0.11 and 0.13. Across all separation levels and for both metrics, k-means lies below Ward: it is the geometrically more

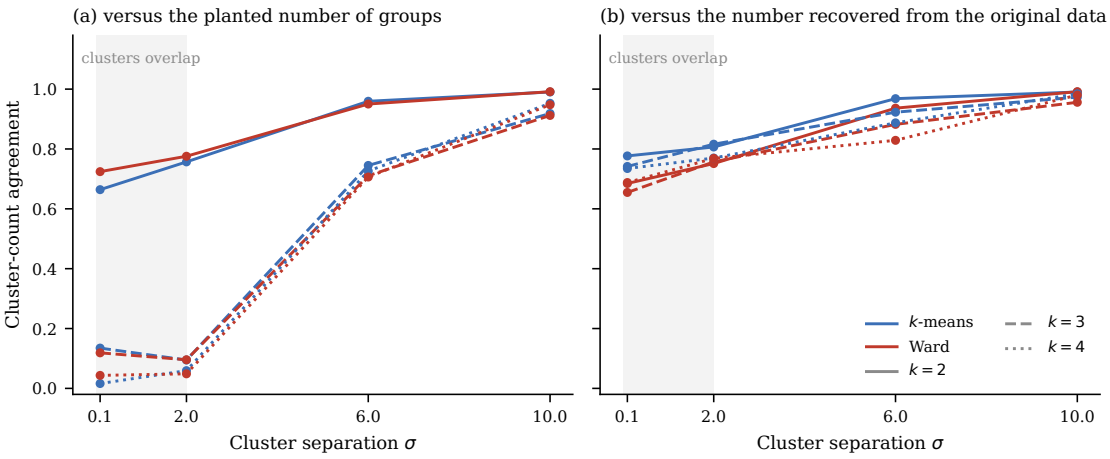


Fig. 3 Cluster-count agreement between synthetic and original data, against separation and stratified by the true number of groups, under two reference conventions. (a) Agreement with the number of groups planted by the simulation (RR). (b) Agreement with the number a clustering algorithm recovers from the original data (AG). In panel (a), agreement is low while the groups overlap (shaded) for $k \geq 3$ and moderate for $k = 2$; panel (b) stays high throughout and depends far less on k, because in the overlapping regime the algorithms often fail to recover the planted count from the original data as well, so both sides of the comparison deviate from the truth in the same way. The two algorithms track each other closely under (a) and separate under (b). Both panels use all 720,000 original–synthetic pairs.

conservative estimator under synthesis. Over the 144 scenarios, k-means has the lower MCD in 133 and the lower MVD in all 144 (paired Wilcoxon $p < 10^{-20}$ for both). The intervals overlap at $\sigma \geq 6$, where both methods yield values close to zero.

Total dispersion distortion grows with dimensionality. Figure 5(a) shows MVD rising with the number of variables: for k-means from 0.06 (95% interval 0.05–0.08) at $p = 2$ to 0.39 (0.32–0.47) at $p = 10$, and for Ward from 0.13 to 0.65. Because v_j in Eq. (7) is a sum of squared deviations over the p coordinates, this growth should be read per coordinate: MVD/p is roughly constant (k-means 0.032, 0.040 and 0.039 at $p = 2, 5, 10$; Ward 0.067, 0.078 and 0.065). Each synthesised coordinate therefore contributes a similar amount of dispersion distortion, and the total accumulates

Table 2 Agreement in the number of groups by design factor, over the full 720,000-evaluation design, under the two references of Eq. (3): the planted number of groups (RR) and the number estimated from the original data (AG). Each entry is a marginal mean over all other factors. Candidate range $\{2, \ldots, \min(k+3, 8)\}$; the fixed-range analysis is in Supplementary Table S2.

		Planted k (RR)		Original $\hat{k}_R$ (AG)	
Factor	Level	k-means	Ward	k-means	Ward
Separation σ	0.1	0.272	0.296	0.751	0.676
	2	0.304	0.307	0.797	0.759
	6	0.810	0.789	0.926	0.882
	10	0.954	0.950	0.983	0.976
Number of groups k	2	0.843	0.860	0.885	0.841
	3	0.473	0.459	0.863	0.812
	4	0.439	0.437	0.844	0.816
Dimensionality p	2	0.616	0.623	0.843	0.770
	5	0.561	0.569	0.866	0.819
	10	0.578	0.564	0.884	0.880
Correlation ρ	0	0.571	0.578	0.800	0.763
	0.4	0.599	0.592	0.929	0.884
Distribution F	Normal	0.604	0.610	0.896	0.898
	Gamma	0.565	0.561	0.833	0.749
Overall		0.585	0.585	0.864	0.823

with p as the axis-aligned leaf structure of Eq. (2) is repeated in every dimension. Unlike the count, which is protected at high p by the collapse of $\hat{k}$ onto small values (Section 3.2), the within-cluster dispersion is measured at the planted k and so exposes this accumulation directly.

Figure 5(b) shows the group-balance error ΔG by distribution family. Both algorithms preserve cluster sizes better on Normal than on Gamma data (k-means 0.028 versus 0.032; Ward 0.057 versus 0.065), consistent with the block artefact being more disruptive on skewed groups. More striking is that k-means yields systematically

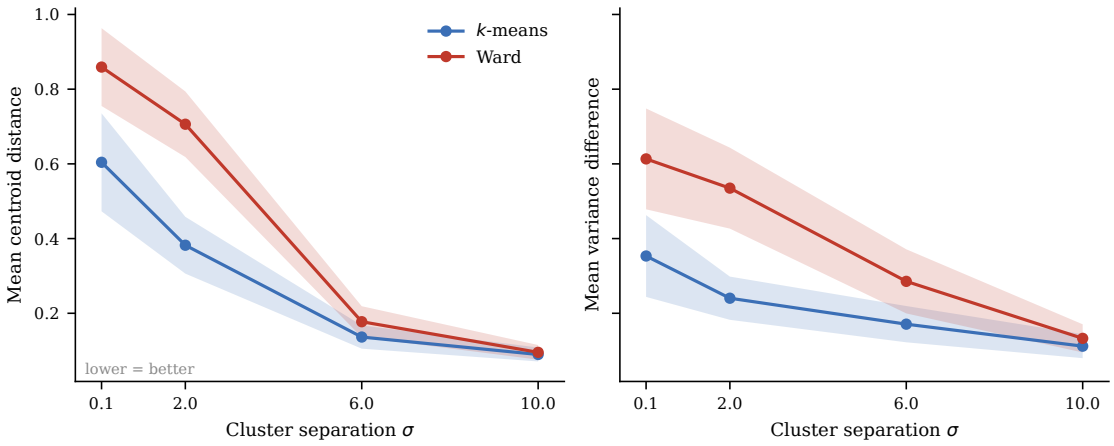


Fig. 4 Geometric fidelity against separation. Points show the mean of the 36 scenario-level means at each separation level; bands are 95% intervals computed across those scenarios. Left: mean centroid distance. Right: mean variance difference. Both panels share the vertical scale. Both metrics decline as groups separate, and k-means remains below Ward throughout.

lower and less variable ΔG than Ward for both distributions (lower in 138 of 144 scenarios; overall 0.030 versus 0.061). Because the planted groups have equal sizes, this difference does not reflect the preservation of a genuine imbalance; it shows that the cluster sizes of a k-means partition fluctuate less under synthesis. This is consistent with the known tendency of the k-means criterion to produce clusters of similar size (Xiong et al., 2009). Ward's linkage uses a related sum-of-squares criterion, but its greedy, irreversible merges let local density irregularities created by the block artefact move whole subsets of points between clusters. Whether k-means would also understate a genuine size imbalance cannot be tested with balanced groups, and is left to future work (Section 4.3).

3.4 Distribution family

The distribution family is the factor that exposes the CART artefact most directly. Figure 6 stratifies count agreement by distribution and separation under both references, and the two panels tell different stories. Against the planted number of

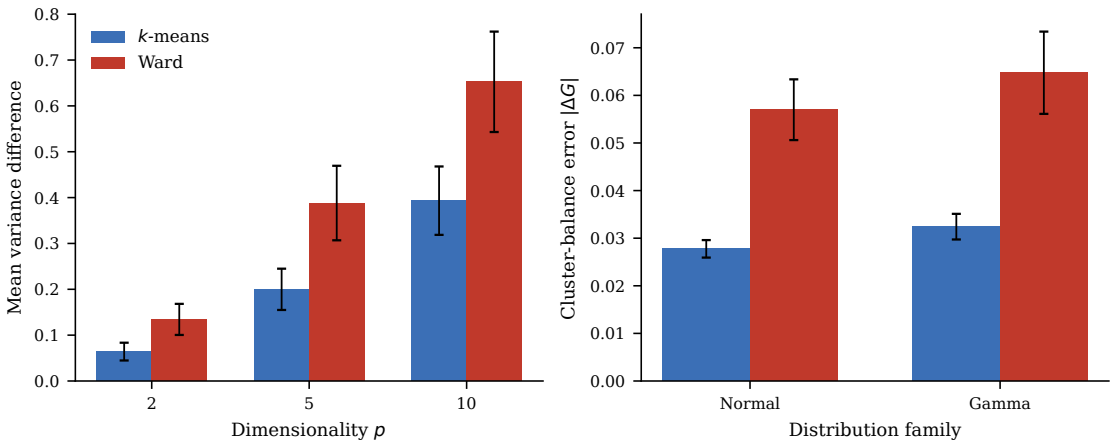


Fig. 5 (a) Mean variance difference by dimensionality: total distortion accumulates as dimensions are added, more so for Ward; per coordinate it is roughly constant (see text). (b) Group-balance error ΔG by distribution family: k-means produces lower and less variable balance error for both distributions. Bars are means of scenario-level means; error bars are 95% intervals computed across scenarios.

groups (panel a), the two families show similar recovery while the groups overlap and diverge at intermediate separation: at $\sigma = 6$, k-means recovers the planted count 86.2% of the time for Normal data but only 75.9% for Gamma data (Ward: 84.6% versus 73.3%). Almost all of this gap, however, is already present in the original data, where the planted count is recovered in 90.0% versus 80.0% of datasets (k-means; Ward 91.1% versus 78.9%). It therefore reflects mainly how difficult skewed groups are to cluster, not what synthesis does to them.

Against the original data (panel b), which isolates synthesis, the Gamma penalty is concentrated where the groups overlap and is much larger for Ward. At $\sigma = 0.1$, Ward's agreement is 0.806 for Normal but only 0.545 for Gamma data, and at $\sigma = 2$ it is 0.878 versus 0.639; the corresponding values for k-means are 0.765 versus 0.737 and 0.888 versus 0.707. At $\sigma = 6$ the gap narrows (k-means 0.947 versus 0.906; Ward 0.924 versus 0.841), and by $\sigma = 10$ it has closed, as separation overwhelms the artefact.

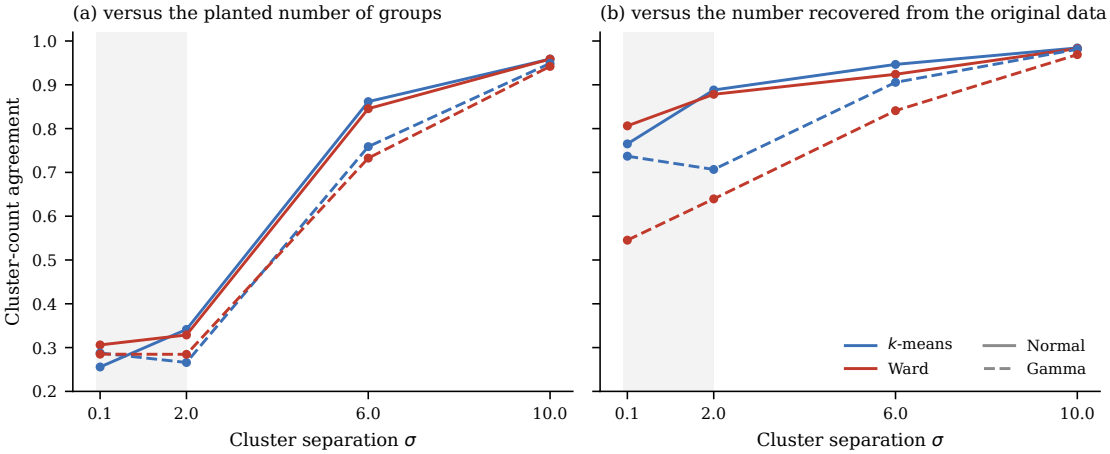


Fig. 6 Cluster-count agreement by distribution family and separation, under the two references. (a) Versus the planted number of groups: the two families show similar recovery while groups overlap and diverge at intermediate separation, where recovery is lower for Gamma data for both algorithms. (b) Versus the number recovered from the original data: the Gamma penalty attributable to synthesis is largest in the overlapping regime and for Ward.

This addresses the first objective. CART preserves the structure needed to recover the number of groups well on Normal data, but its hardening of skewed densities is associated with lower agreement on Gamma data, most markedly when groups are not well separated and for the hierarchical method.

3.5 Agreement between the two algorithms

Whether the two methods differ in robustness to synthetic distortion was the third objective, and the answer depends on the reference. Against the planted number of groups the two are indistinguishable: overall recovery is 0.585 for both, a paired Wilcoxon signed-rank test across the 144 scenario-level means finds no significant difference ($W = 3300$, $p = 0.39$), and the two algorithms reach the same outcome in 92.0% of individual evaluations. Against the original data, which isolates synthesis, k-means is more stable: overall agreement is 0.864 versus 0.823, k-means is higher in 95 of the 144 scenarios (24 ties), and the paired test is significant ($W = 1665$, $p <$

0.001). The conclusion is unchanged under the fixed candidate range (Supplementary Section S3). Together with the geometric results, this indicates that k-means is the more robust of the two methods to CART synthesis, whereas the prior expectation that a hierarchical method would be more robust is not supported.

As a complementary check on internal cohesion, the difference in mean silhouette width between synthetic and original partitions is small and centred near zero at every separation (within ± 0.012). The absolute levels are nearly identical (k-means 0.396 versus 0.397; Ward 0.376 versus 0.377), suggesting that the apparent clustering quality is similar before and after synthesis. This is reassuring, but it also means that silhouette alone would fail to flag the geometric distortions the other metrics detect, an instructive negative result for anyone tempted to validate a synthetic release with an internal index.

4 Discussion

This study evaluated whether CART-generated synthetic data preserve the structural properties that cluster analysis relies on, comparing k-means and Ward's hierarchical clustering across a 144-scenario factorial design and 720,000 synthetic-versus-original evaluations. Four findings emerge.

First, separation dominates agreement in the number of groups. Agreement with the original data rises from 0.75 (k-means) when the groups overlap to 0.98 when they are well separated, and recovery of the planted count rises from roughly 0.27 to above 0.95. The number of groups strongly affects recovery of the planted count (higher at $k = 2$, 0.84, and lower at $k \geq 3$, below 0.48) but has little effect on agreement between original and synthetic estimates. Overall, recovery is highest for simpler cluster structures, with well-separated and fewer groups, and lowest for more complex structures, with overlap and a larger number of groups; most of the effect of k, however, is a property of the clustering problem that synthesis reproduces rather than a distortion that synthesis introduces.

Second, we found no significant difference between the two algorithms in recovery of the planted count (0.585 versus 0.585; $p = 0.39$), but k-means is the more stable of the two when synthetic estimates are compared with those from the original data (0.864 versus 0.823; $p < 0.001$). The same ordering holds for the geometry: k-means shows lower centroid and variance distortion at every separation and smaller fluctuations in cluster sizes. For an applied analyst, the practical implication is that the choice of algorithm matters more for the fidelity of a synthetic-data analysis than accuracy against a ground truth would suggest. It also raises a question that this design cannot answer: because the k-means criterion favours clusters of similar size, a k-means analysis of synthetic data may understate a genuine imbalance, which only a design with unequal group sizes can test.

Third, CART is not utility-neutral, and its performance depends on the underlying distribution. On Normal data it is close to neutral; on skewed Gamma data it hardens smooth density gradients into axis-aligned blocks, visibly distorting geometry and being associated with lower agreement, particularly for Ward's linkage and when groups overlap. Total dispersion distortion grows with dimensionality, because each synthesised coordinate contributes a similar amount of distortion.

Fourth, and methodologically, an internal validity index is not a sufficient check. Silhouette values remain almost unchanged even where centroid and dispersion fidelity degrade substantially. A practitioner who validates a synthetic release by confirming that its silhouette profile matches the original may conclude that structure is intact in precisely the settings where this analysis shows it is not. A related methodological point concerns the reference: the planted truth, the natural choice in a simulation, answers a different question from the one a data holder faces, and in this study it masked the difference between the two algorithms.

An interesting corollary of the design is that clustering can be read in the other direction, as an instrument rather than an object of study. Because a partition responds to local density, separability and dispersion simultaneously, the agreement

between partitions recovered from original and synthetic data functions as a sensitive summary of geometric resemblance, one that reacts to distortions which marginal and bivariate checks may not detect. We did not pursue this here, but the metrics of Section 2.4 could be repurposed as general-utility diagnostics rather than task-specific ones.

4.1 Relation to previous work

The study extends Chen (2025) in three ways: a structurally distinct second algorithm, a skewed distribution family that exposes tree-synthesis artefacts, and structural metrics that compare partitions recovered from the original and synthetic data. The conclusion that CART is a reasonable default for well-behaved data but degrades on skewed distributions is consistent with the broader `synthpop` literature (Nowok et al., 2016; Snoke et al., 2018), and the qualitative block artefact matches the known axis-aligned bias of sequential tree synthesis. The finding that utility rankings depend on the downstream task echoes benchmarking work in the health-records setting (Yan et al., 2022; El Emam et al., 2022), where the same generator changes position depending on which analysis is used to score it, and an applied study of wearable physiological data in mental-health monitoring similarly found that the clustering utility of `synthpop` releases depends on the synthesis settings (Vidiella Villegas and López, 2025). The greater sensitivity of Ward's linkage to perturbation of the data is in line with the classical robustness comparisons of Milligan (1980). A parallel question arises when values are missing rather than synthesised: Aschenbruck, Szepannek, and Wilhelm (2023) evaluate how imputation strategies affect the clusters subsequently recovered from mixed-type data, and reach a structurally similar conclusion, that the imputation model and the clustering method interact and cannot be assessed in isolation. Since CART underlies both sequential synthesis and widely used non-parametric imputation, the two literatures are closer than their vocabularies suggest.

4.2 Limitations

The evaluation uses controlled simulations with known ground truth rather than real data. This isolates mechanisms cleanly but limits direct generalisation to data with missingness, mixed types, unequal group sizes and unknown data-generating processes. The generator is limited to CART with a single parameterisation (minimum leaf size ten, fixed variable order) and the algorithms to k-means and Ward; density-based and model-based clustering, other synthesis settings, and parametric or deep generative synthesisers, may behave differently. Sample size was fixed at $N = 1000$, so small-sample behaviour is unexamined. The number of groups is selected by the silhouette criterion over $\hat{k} \geq 2$, so a single-cluster solution can never be chosen, even when the groups overlap so strongly that one cluster is the most reasonable description; other selection rules (Milligan and Cooper, 1985) may respond differently to synthesis. Finally, count agreement is inflated where the estimator collapses onto a small number of candidate values, as happens at high p and low σ, and should be read together with the geometric metrics.

Two conventions in the definition of MCD and MVD deserve emphasis, because both affect the numbers themselves and not only their interpretation. The first is what the metrics compare. Here they compare the partition recovered from the original data against the partition recovered from the synthetic data, which is the quantity a claim about synthesis requires; the superficially similar practice of comparing a recovered partition against the planted labels of the same dataset measures clustering accuracy instead, and carries no information about the generator. The second is scale. Both metrics are computed after standardising with a scaler fit on the original data and applied unchanged to the synthetic data, so both are expressed in units of the original feature scale. Standardising each dataset separately rescales away part of the distortion the metrics are meant to capture, and omitting standardisation altogether makes MVD track absolute cluster spread (which grows with σ by construction), so that the metric rises with separation when what it is measuring is

the design parameter. Absolute values of MCD and MVD are therefore not comparable across studies that differ in either convention, and any replication should state both explicitly. Both metrics are also unnormalised sums over coordinates, so their absolute values grow with p even at constant per-coordinate distortion; comparisons across dimensionality should use per-coordinate or relative versions.

4.3 Future work

The central lesson generalises beyond the specific methods studied: synthetic data utility is task-relative, and a generator that preserves marginal resemblance need not preserve the higher-order geometry a clustering workflow depends on. Practically, the results argue for validating structural fidelity (distances, dispersion, separability, balance) against the intended analysis, and for treating a clustering comparison as a more demanding resemblance check than marginal summaries can provide. Natural extensions include other synthesisers (parametric mixtures, copulas, deep generative models) and other CART settings, non-convex and density-based clustering, unequal group sizes, varying N to probe small-sample behaviour, and chance-corrected partition comparisons such as the adjusted Rand index (Hubert and Arabie, 1985) computed on a common set of points. A promising direction is to model the discrepancy Δ of Eq. (1) directly as a learned function of the design factors, turning the utility surface itself into a predictive tool.

As a concrete next step, we propose applying the framework to the wearable dataset described in Vidiella Villegas and López (2025), to which we have direct access. Beyond providing a real-data test of the present design, this dataset would allow a comparison that neither study can perform alone. The present work evaluates synthesis as substitution, in which the synthetic dataset replaces the original. The applied study instead evaluates augmentation, in which the synthetic records are appended to the real ones and the union is clustered. These are different operations, and their effects are currently confounded with sample size: doubling the number of

observations alters neighbourhood parameters and silhouette values independently of any distortion introduced by the generator. A three-way comparison of the partitions recovered from the real data, from the synthetic data alone, and from the augmented set, evaluated with the same metrics used here, would separate the effect of the generator from the effect of the additional sample. It would also make it possible to test whether the parameterisation of the generator, rather than the generator itself, accounts for the block artefact documented in this paper, since the applied study reports a substantial improvement in utility after replacing CART with conditional inference trees (Hothorn, Hornik, and Zeileis, 2006) and increasing the minimum leaf size. We regard this comparison as a natural continuation of the present study and as a practical way of establishing which of its conclusions generalise beyond the simulated setting.

5 Conclusions

In our view, the main contribution of this study is that its discrepancy framework does not depend on the simulation that illustrates it. The quantity $\Delta(\theta_R, \theta_S)$ is defined for any reference dataset and any generator, and the five factors varied here all have direct empirical counterparts: separation corresponds to how distinct the underlying groups are, the number of groups to the number of latent subpopulations, dimensionality to the number of recorded variables, correlation to the dependence between features, and the distribution family to the shape of the observed densities. The simulation fixes these factors by construction, but in an applied dataset they can be measured. This means the framework can be transferred to real data without modification, and we believe such a transfer would be informative rather than merely confirmatory. Real datasets occupy regions of the design space that the present grid does not reach. Behavioural and physiological measurements routinely show correlations above 0.95 between related features, well beyond the maximum of $\rho = 0.4$ examined here, and skewness values considerably higher than those produced by

the Gamma family. They also exhibit properties the simulation excludes altogether, such as strongly unbalanced group sizes, ambiguous missingness in which a zero may indicate either genuine inactivity or a non-worn device, and heterogeneity between individuals. Each of these is a plausible source of additional distortion under sequential tree synthesis, and each would help to refine the conclusions reported here. We consider the unbalanced case particularly relevant, because the group-balance error ΔG and the hypothesis that the k-means criterion understates imbalance can only be fully tested when a substantial imbalance is present in the reference data.

Within the region of the design space studied here, synthetic data generated with CART can reproduce the appearance of clustering structure while subtly reshaping its geometry. Both algorithms recover well-separated structure reliably and struggle together when groups overlap; where they diverge is in how faithfully each reproduces, on synthetic data, the partition it would have found on the original data, and in the geometric bias each imprints on the recovered partition. The utility of synthetic data for clustering is therefore best understood not as a property of the data but as a property of the data–generator–algorithm triple, to be verified rather than assumed.

Supplementary information. The online supplementary material contains the recovery heatmap by number of groups and separation, the contingency table of original versus synthetic estimates, the sensitivity analysis for the candidate range of $\hat{k}$, and the independent-resampling baseline.

Declarations

Funding: [To be completed by the authors.]

Competing interests: [To be completed by the authors.]

Data availability: All data analysed in this study are simulated; they can be regenerated exactly from the configuration file and seeds distributed with the code.

Code availability: The full pipeline, from data generation through to the figures and tables reported here, is available at https://github.com/Sterrysx/synthetic-clustering. Every stage is driven from a single configuration file and can be re-executed end to end.

Author contributions: [To be completed by the authors.]

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Appendix A Clustering objectives

k-means partitions N observations $\{x_i\}_{i=1}^N$, $x_i \in \mathbb{R}^p$, into k non-overlapping groups by minimising the within-cluster sum of squares (MacQueen, 1967). Writing $\mathcal{C} = \{C_1, \ldots, C_k\}$ for the partition and μ_j for the centroid of C_j,

$$\arg\min_{\mathcal{C}} \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2, \qquad \mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i. \tag{A1}$$

Lloyd's iteration alternates assignment and update steps until the partition stabilises, converging to a local optimum.

Hierarchical clustering builds a dendrogram by successive merging (Ward, 1963). Each observation begins as a singleton; at every step the two clusters whose merger least increases the total within-cluster error sum of squares are combined. With

$$\mathrm{ESS} = \sum_{j} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2, \qquad \Delta_{\mathrm{Ward}}(C_a, C_b) = \frac{|C_a|\,|C_b|}{|C_a| + |C_b|} \|\mu_a - \mu_b\|^2, \tag{A2}$$

a flat partition into k groups is obtained by cutting the dendrogram at the corresponding height.

Appendix B Computational complexity and cost

This appendix states what the study costs to run and why. Throughout, N is the sample size, p the dimensionality, k the number of groups, I the number of Lloyd iterations to convergence, $K = 7$ the number of candidate values of $\hat{k}$, and $A = 2$ the number of algorithms.

B.1 Cost of the components

Table B1 collects the asymptotic cost of each component together with the number of times the executed design invokes it. The invocation counts follow from the design: 144 scenarios $\times$ $n = 5$ replicates $= 720$ units, each with $m = 1000$ synthetic draws, giving 720,000 original–synthetic pairs.

Table B1 Cost per invocation and invocation count. Exact minimisation of the k-means objective NP-hard in general and remains so for $k = 2$ in arbitrary dimension (Inaba, Katoh, and Imai, 1994) Lloyd's local-search heuristic (Lloyd, 1982) is used, restarted ten times.

Component	Cost per invocation	Inv
CART synthesis (Breiman et al., 1984)	$O(p^2 N \log N)$	1
Lloyd iteration (MacQueen, 1967; Lloyd, 1982)	$O(IkNp)$, $\times 10$ restarts	5
Ward linkage (Ward, 1963; Murtagh and Contreras, 2017)	$O(N^2 p)$ time, $O(N^2)$ space	5
Silhouette (Rousseeuw, 1987)	$O(N^2 p)$	1
Hungarian alignment (Kuhn, 1955; Jonker and Volgenant, 1987)	$O(k^3)$	1

Two structural facts follow. First, the *binding* term is $O(N^2 p)$, contributed by Ward's linkage and by the silhouette, and it is a property of the sample size rather than of the design: doubling N quadruples the work, whereas doubling m or the number of scenarios only doubles it. The $O(N^2)$ working set of the linkage is also what caps N in practice, since each worker holds its own distance structure; at $N = 1000$ this is roughly 8 MB per worker and irrelevant, at $N = 10^4$ it is 800 MB and binding. Second, model selection dominates the fit count: estimating $\hat{k}$ costs K fits and K silhouette evaluations per algorithm per dataset, so the fidelity stage

performs $720 \times A(K{+}1) + 720{,}000 \times A(K{+}1) = 1.15 \times 10^7$ clustering fits rather than the 1.44×10^6 that fitting at a single known k would require. Where a cheaper formulation exists we did not pursue it: Ward admits an $O(N^2)$ nearest-neighbour-chain implementation (Murtagh and Contreras, 2017; Müllner, 2011) and k-means can be seeded to bound its expected error (Arthur and Vassilvitskii, 2007), so the figures below are conservative relative to a tuned implementation.

One algorithmic choice does change the constant materially. The partition of each original dataset is computed once per replicate and reused across that replicate's 1000 draws. Computing it inside the draw loop would multiply the original-side work by 10^3 and produce identical output, since the original data do not vary within a replicate.

B.2 Parallel execution

The design is a set of independent units, so the computation is embarrassingly parallel in the strict sense: no communication, no shared mutable state, and a serial fraction confined to reading configuration and writing the merged result. Amdahl's law is therefore not the operative constraint at this scale; hardware contention is, and the implementation targets three specific sources of it.

Thread oversubscription. The clustering routines call into BLAS, which by default spawns one thread per core *inside each worker*. With P worker processes this creates $P \times C$ runnable threads on C cores, and the resulting context-switching and cache thrashing dominate the useful work. All three stages therefore pin `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS` and `MKL_NUM_THREADS` to 1 before importing any numerical library, leaving parallelism entirely to the process pool. The effect is not marginal: on the reference machine, 18 concurrent workers each running an identical model-selection scan complete in 1.9 s with threads pinned and 84.5 s without, a factor of 44.

Load imbalance. Unit cost varies with p and k, so a static partition of the task list leaves workers idle at the tail. The synthesis stage instead uses a work-stealing queue: tasks are serialised into a spool directory and each worker claims one by `rename(2)` into a *doing* directory. Rename within a filesystem is atomic, so exactly one worker can win a given task and no lock is required. Workers pull candidates in batches of 200 rather than listing the spool per task, which at 1.44×10^5 tasks reduces directory-scan overhead from hours to seconds. The Python stages achieve the same effect through a dynamically scheduled pool.

Restart cost. Each unit checkpoints its own result on completion, so an interrupted run resumes at the last completed unit rather than from the start. At a stage cost of several hours this converts a failure from a total loss into the loss of the units in flight.

Figure B1 shows the measured strong-scaling curve. The shape is the expected one for a compute-bound, memory-light workload on a simultaneous multithreading processor, and it makes the operating point explicit: 18 workers buy a 10.0× speed-up over serial execution, while 12 would give 8.8× at much better efficiency. Neither choice is a contribution; the point of measuring is that the default of one worker per hardware thread is not obviously right, and here it is not.

B.3 Measured cost

The synthetic datasets occupy 21 GB as uncompressed Parquet. Cost is linear in m: the same design at $m = 100$ completes in about a tenth of the time and reproduces every estimate reported here to within 0.025, so the larger run buys precision rather than a change in any conclusion.

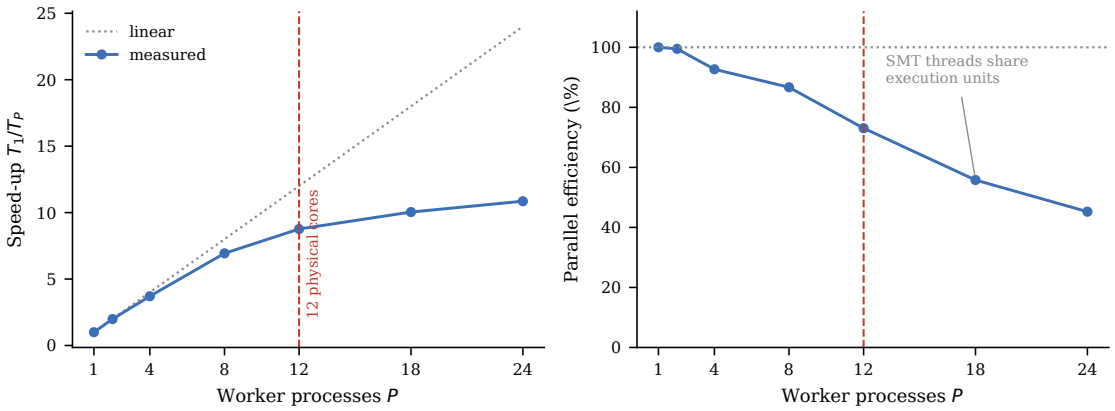


Fig. B1 Strong scaling of the model-selection workload on the reference machine, 72 independent units, BLAS pinned to one thread per worker. Scaling is near-linear to 8 workers (6.9×, 87% efficiency) and the knee falls at 12, the physical core count. Beyond it the additional workers occupy SMT siblings that share execution units with an already-saturated core, so throughput still rises but efficiency falls to 56% at 18 workers and 45% at 24. The production runs use 18: past the efficiency knee, but chosen for wall time on an otherwise idle machine. The efficiency figure is not a curiosity: at 56%, the clustering stage costs 3.6 h rather than the 2.0 h that dividing its serial cost by 18 would suggest.

Table B2 Wall-clock cost at $m = 1000$ on an AMD Ryzen 9 9900X (12 cores, 24 threads, 46 GiB) with 18 workers. The evidence differs by row and is stated rather than blurred: the fidelity stage is instrumented; synthesis is the interval over which its 1.44×10^5 outputs were written, so it includes any idle time inside that interval; the clustering stage was not instrumented, so its entry is a measurement on a uniform random sample of 3000 of its 1.44×10^5 inputs, run at the same worker count, scaled to the full set; this is consistent with the loose upper bound of 5.6 h implied by its output timestamps.

Stage	Evidence	Wall time	Unit cost
Synthesis (1.44×10^5 datasets)	write interval	1.24 h	31 ms/dataset
Clustering (7.2×10^5 pairs)	sampled	$\approx$ 3.6 h	$\approx$ 18 ms/pair
Fidelity (1.15×10^7 fits)	instrumented	4.44 h	1.4 ms/fit
Total		$\approx$ 9.3 h	